The rain is burning! eles-0065

1. Explain how acid rain is formed.

2. What gases cause the formation of acid rain?

3. What are the most common sources of the gases that cause acid rain?

4. (a) Describe what the pH scale is.

   (b) What is the pH of normal rainwater?

   (c) What is the pH of acid rain?

   (d) When a liquid becomes more acidic, does its pH increase or decrease?

5. Describe what happens when metal comes into contact with an acid.

6. Many statues in countries such as Greece and Italy have survived intact for thousands of years. However, in the last hundred years or so, they have started to crumble and break down. Explain why this is happening after so many years of being unaffected.

7. Deciduous trees drop their leaves in autumn and grow new ones in spring. Evergreen trees are always leafy, regenerating leaves all year round. Which type of tree do you think would be more affected by acid rain? Explain your answer.

8. Describe at least three ways in which we can reduce the incidence of acid rain.

9. Acid rain does not always fall in the areas where there is heavy industry; often it falls hundreds of kilometres from industrial areas. Why do you think this is the case?

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10. Identify at least three effects that acid rain has on the environment.

11. Throughout history, what we now call acid rain was recorded long before there were factories, cars or power stations. These instances of acid rain occurred after violent volcanic eruptions. Explain how volcanic eruptions could cause acid rain.

12. Describe what chemical scrubbers are and where they may be used.

13. Explain how acid rain affects water systems.

14. Australia does not have as much of a problem with acid rain as North America and Europe. Propose an explanation for this.

15. Most of our electricity comes from power stations that burn fossil fuels. Name at least two alternative energy sources that could be used to provide electricity.

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